Module 1 Practicum

By: Andy Steiner

Section 1: Brain Area Report: Claustrum

Subsection 0: In Section 1, I will explore the claustrum's structural and functional characteristics, including its connectivity with various cortical regions, genetic expression patterns, and its proposed role in cognitive processes such as attention and network regulation. By examining both anatomical and functional data, I aim to highlight the claustrum's potential as a key integrative hub in the brain.

Subsection 1: Claustrum Structure and Functional Connectivity

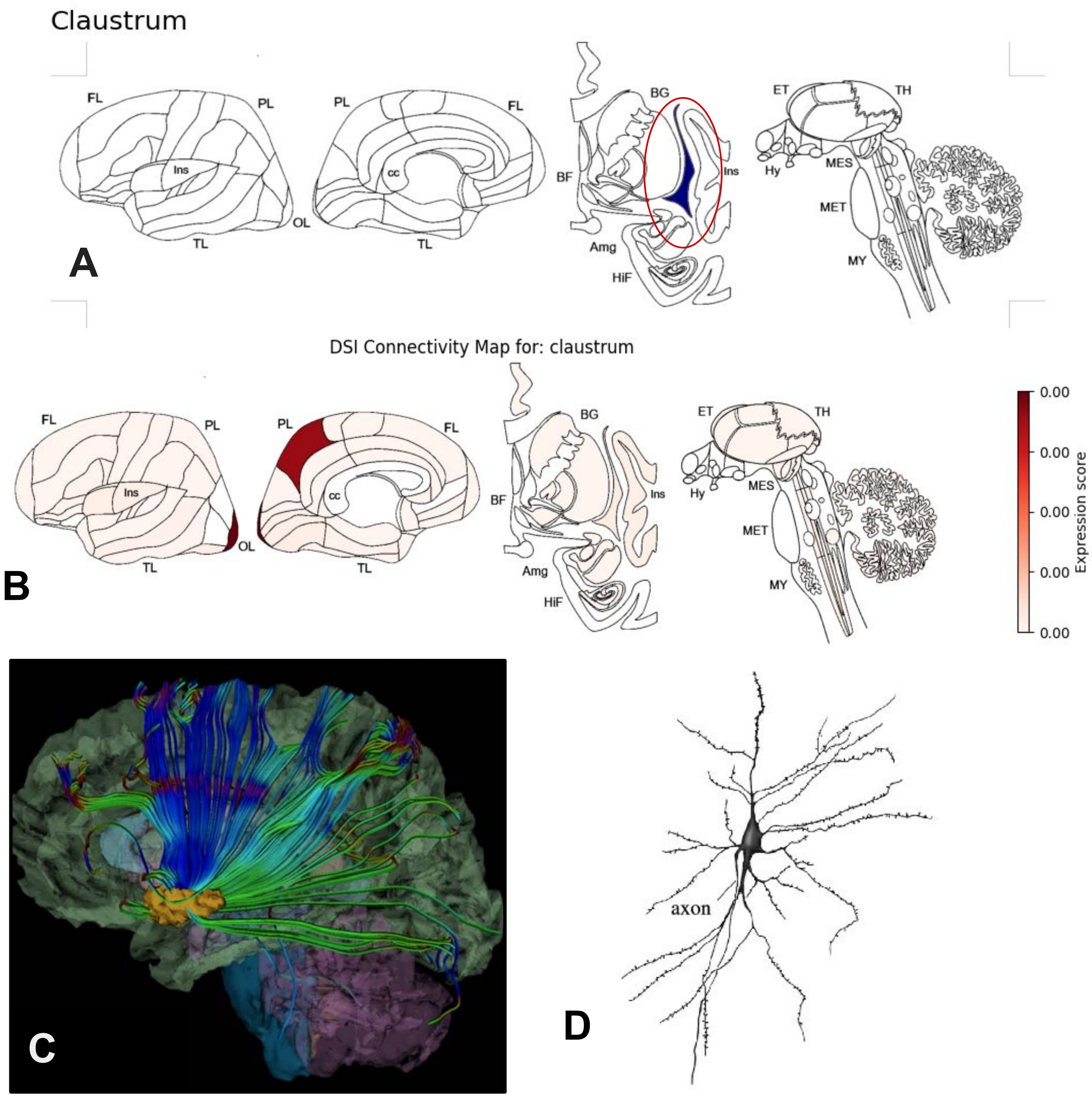


Figure 1. The claustrum is a thin sheet of grey matter located deep in the brain between the insular cortex and the putamen. It is thought to play a role in integrating information relating to consciousness and attention.

(A) The claustrum is shown in blue (circled red). It is located in the basolateral telencephalon lateral to the putamen and medial to the insular cortex. **(B)** DSI connectivity for the claustrum is shown. The highest connectivity, the precuneus and occipital pole, are highlighted dark red. **(C)** A white matter fiber tractography (DTI) map where fibers linking the claustrum to other brain regions are shown in blue (dorsal-ventral), and green (anterior-posterior) is shown (Torgerson et al., 2014). Due to the strong connection between the claustrum and frontal, parietal, occipital lobes, as well as the precuneus and anterior cingulate cortex (ACC), it is hypothesized that the claustrum is involved in regulating and switching between multiple cortical networks including the frontoparietal (FPN), salience (SN), and default mode network (DMN). This is further supported by that fact that the claustrum receives input from the frontal cortex and projects back to the occipital lobe (Madden et al., 2022). **(D)** Shown is the most abundant type of neuron in the claustrum, the Golgi Type 1 neuron. The predominant cell type is the glutamatergic excitatory neuron, although there is a high expression of parvalbumin (PV) inhibitory interneurons towards the center of the claustrum (Bjerke et al., 2021).

Subsection 2: Clastrum Genetic Expression and the ADRA2C gene

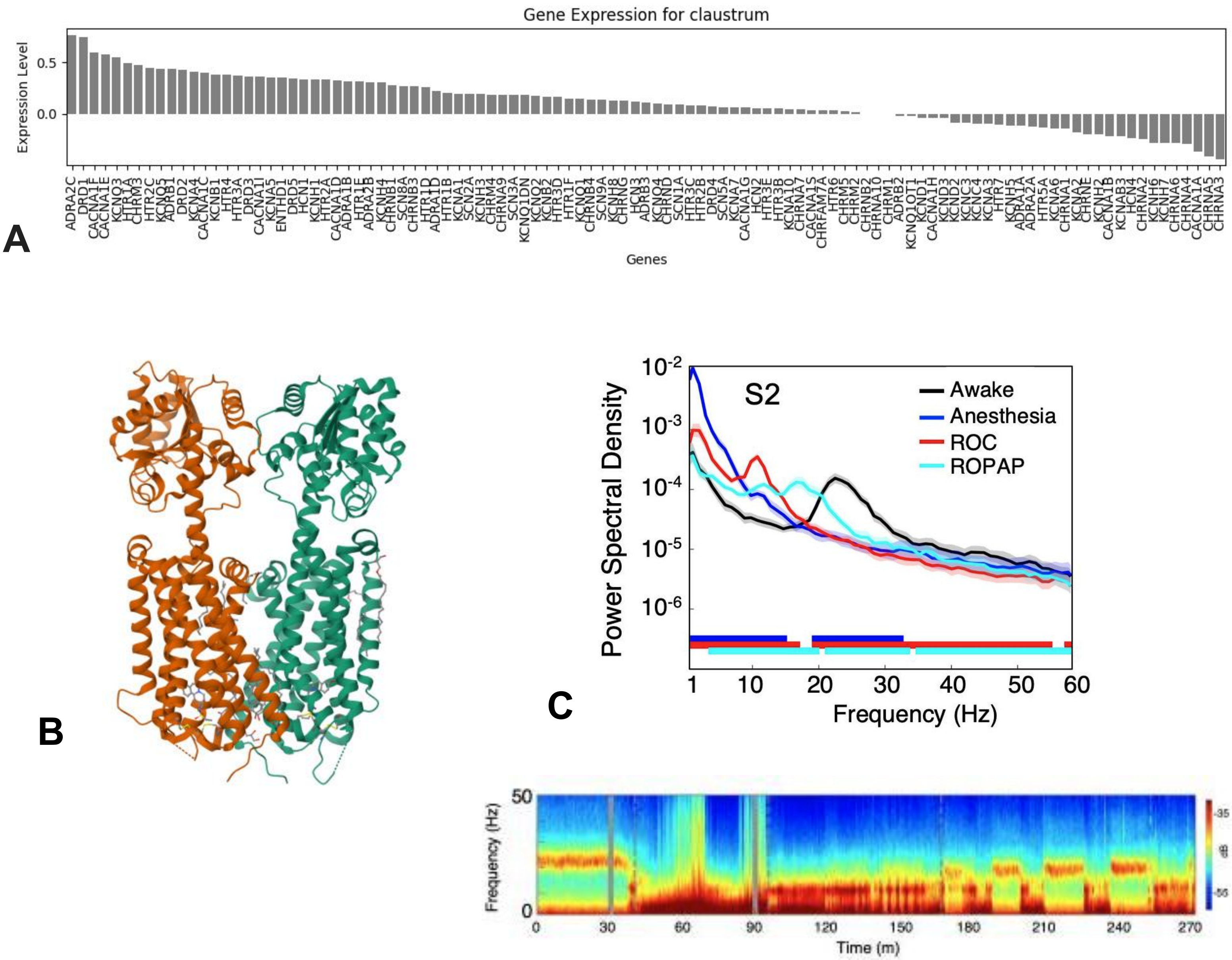


Figure 2. The clasutrum contains a mix of excitatory and inhibitory neurons. A highly expressed receptor in the claustrum (ADRA2C), is thought to play a role in modulating consciousness and conscious states.

(A) Genetic expression for the claustrum is shown. ADRA2C is the highest expressed gene, and CHRNA3 is the most suppressed gene. ADRA2C encodes the alpha-2C adrenergic receptor, which bind to epinephrine and norepinephrine. Diseases associated with this gene include Raynaud Disease and Attention Deficit-Hyperactivity Disorder (ADHD). CHRNA3 encodes the alpha-3 nicotinic acetylcholine receptor. **(B)** The crystal structure of human alpha-2C adrenergic G protein-coupled receptor is shown. ADRA2C appears to be involved in modulating neurotrasmission at lower levels of nerve activity, while another subtype (ADRA2A), inhibits release at higher frequencies (Hein et al., 1999). **(C)** This figure shows local field potential (LFP) time-domain spectrograms (bottom) and the corresponding traced averages (top). The study investigated the α2-adrenergic receptor's role in altering conscious states during anesthetic periods. Dexmedetomidine is a highly selective agonist, and is unique to most currently available anesthetics as others are known to act on multiple receptors. The study showed that neural changes during α2-adrenergic agonist dexmedetomidine-induced altered states of consciousness are consistent with propofol-induced loss of consciousness (LOC) and return of consciousness (ROC) in non-human primates (Ballesteros et al., 2020).

Subsection 3: The Claustrum's Role in Cognition

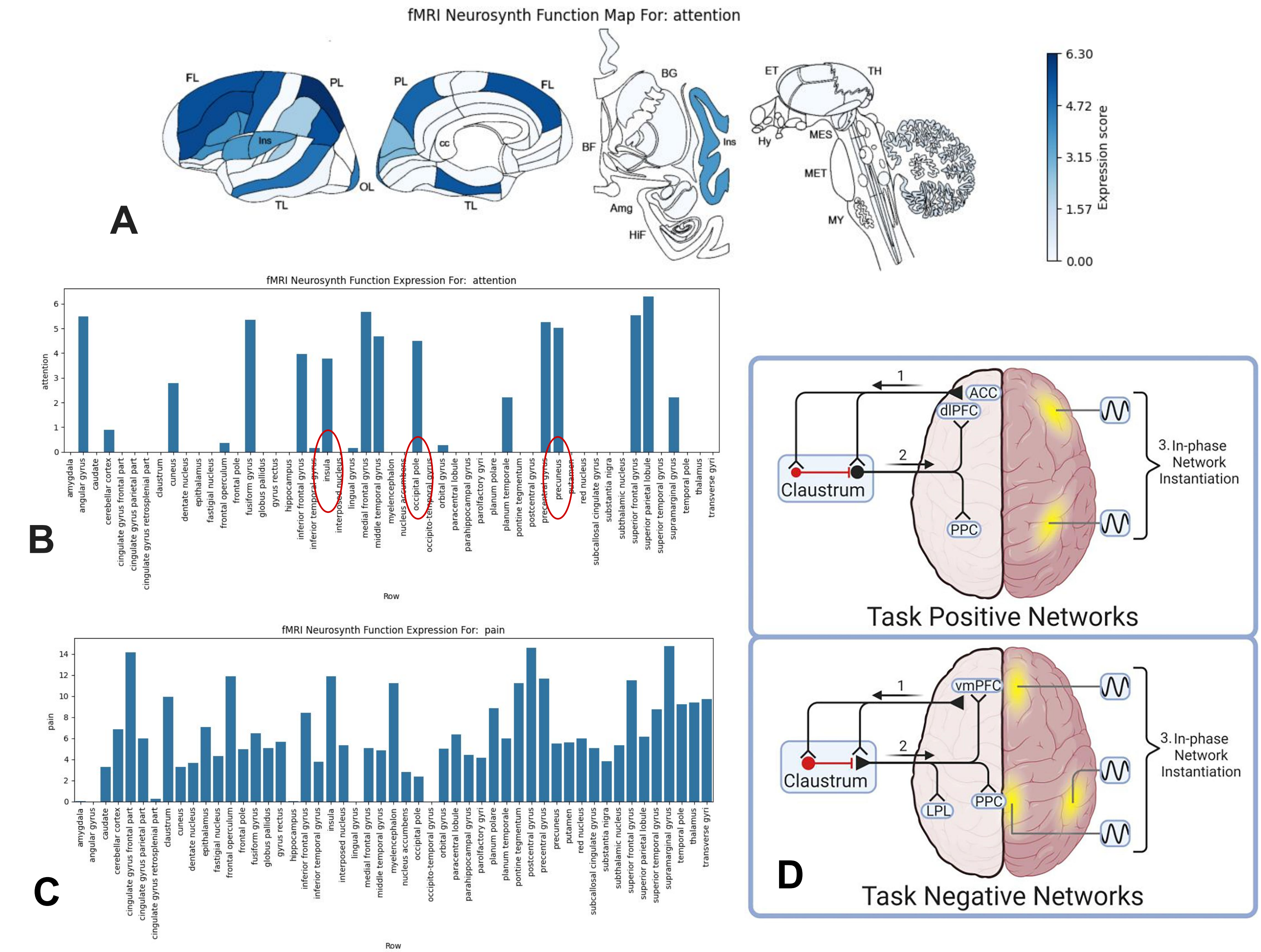


Figure 3. The claustrum plays a complex role in cognition, and is challenging to study due to its irregular shape and small size.

(A) The fMRI neurosynth function expression for attention is shown. This image is included even though there is no visible claustrum activity, as it outlines one of the challenges of using fMRI to study the claustrum. The voxel size of a standard fMRI is very large compared to the claustrum, which results in contamination from nearby regions and difficulty picking up signals. It is understood that the claustrum plays a crucial role in initiating attention and modulating information, not necessarily attention itself which could also explain this trend. Despite not showing any activity in the neurosynth data, neighboring regions like the insula, as well as areas that the claustrum is heavily connected to like the precuneus and occipital pole are active. In order to get more accurate results, "Small Region Confound Correction" (SRCC) or other methods, like seed-based network analysis are preferred. **(B)** Shown is the accompanying bar graph for the attention neurosynth map. The insula, precuneus, and occipital pole and circled red.

(C) The fMRI neurosynth function for pain is shown for comparison. Although claustrum activity is present, widespread activation across nearly all brain regions highlights a key challenge in studying the claustrum: obtaining accurate and specific recording data. **(D)** A diagram showing a theory for the claustrum's role in cognition, the Network Instantiation in Cognitive Control (NICC) model, is shown. This model proposes that the frontal cortex directs information relating to cognitive control to the claustrum, which then processes this information and relays to modulate other cortical networks. For example, the ACC is proposed to activate claustrum neurons that selectively synchronize components of the task positive network (goal directed and focused tasks). While other areas activate claustrum neurons that synchronize components of task negative network (DMN/during rest). This model supports the idea that a role the claustrum plays in cognition is switching between and regulating multiple cortical networks.

Section 2: FMRI Function Report

Subsection 0: In Section 2, I will examine the default mode network (DMN) using fMRI data, focusing on its activation patterns during rest and its suppression during task-oriented activities. I will explore the functional connectivity within the DMN and its interactions with other networks, such as the salience network, particularly in the context of psychological disorders like ADHD and anxiety. Additionally, I will investigate the genetic and receptor expression profiles associated with key DMN regions to understand the neurochemical basis of its function.

Subsection 1: The Default Mode Network (DMN)

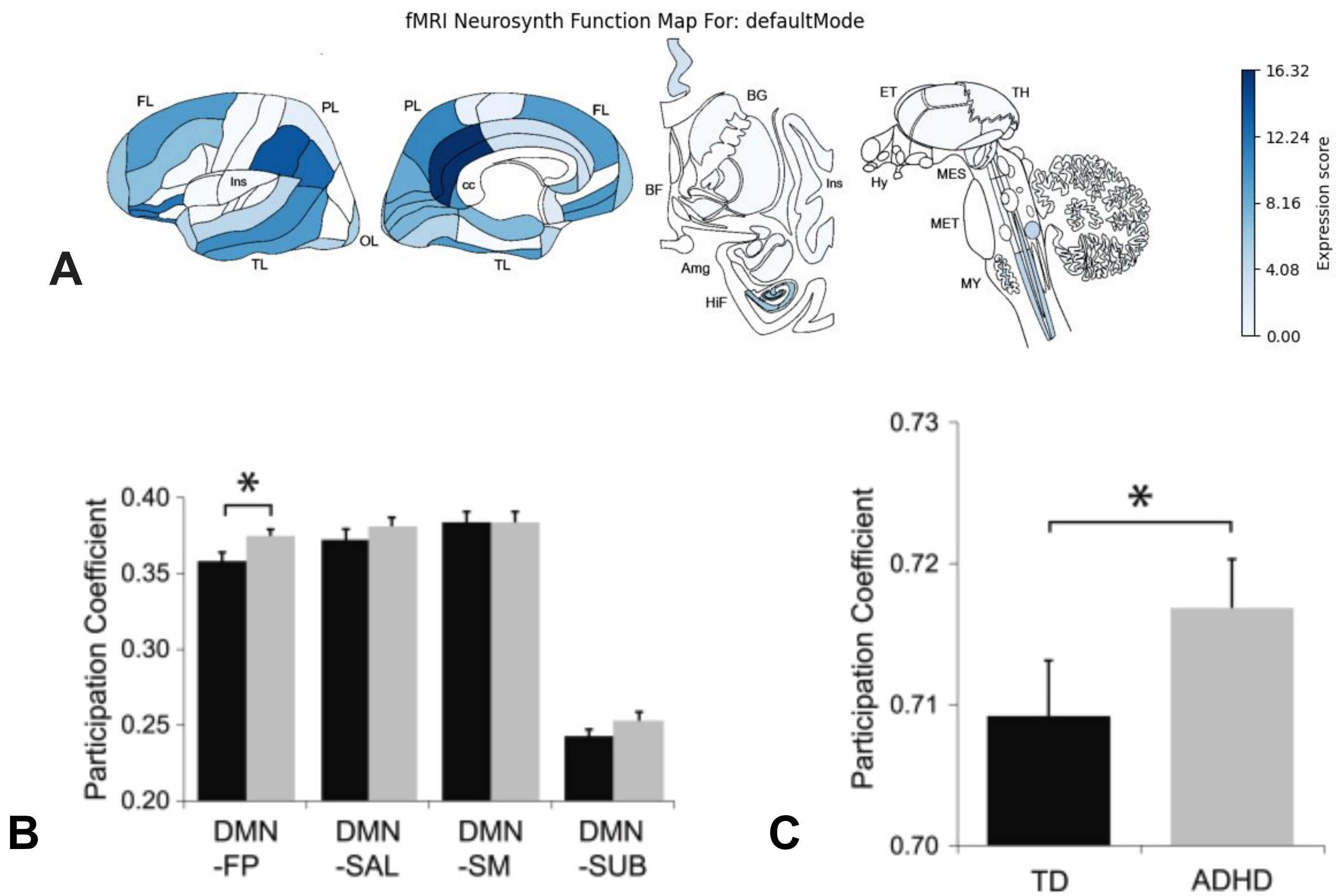


Figure 1. The default mode network (DMN) is a group of brain regions active during rest, which deactivate during focused goal-oriented tasks. The DMN is thought to play a key role in memory, imagination, and maintaining a sense of self, although it's true cognitive function is unclear.

(A) The neurosynth functional map for the default mode network (DMN) is shown. The DMN is best described as the group of brain regions that are active during rest, introspection, and and self-referential thinking (task-negative), but deactivate during focused, goal-directed tasks (task-positive). In the absence of attention to external stimuli, the DMN switches or "defaults" to internally focused thought processes. Researchers are interested in the DMN not only for its activation during rest, but also because *it is suppressed during attention directed tasks*. This suppression of the DMN plays an important role in various psychological disorders. For example, there is difficulty suppressing the DMN in those with ADHD and anxiety (leading to hyperactive DMNs). This is hypothesized to lead to trouble staying focused during goal-directed (task-positive) tasks for ADHD, and increased self reflection/excessive rumination in individuals with generalized anxiety disorder (GAD) (Li et al., 2023).

(B) Using participation coefficients (PC) across five different networks, a metric that measures how connected a node is across networks, a study found a significant difference between the DMN and fronto-parietal (FP) networks, indicating that the DMN is overly connected to task-relevant networks in those with ADHD when compared to typically developing children (TD) (Duffy et al., 2021). **(C)** Using PC across all five networks, the study found that in individuals with ADHD the DMN is overly connected (and hyperactive) compared to TD (Duffy et al., 2021).

Subsection 2: Mapping The Default Mode Network (DMN)

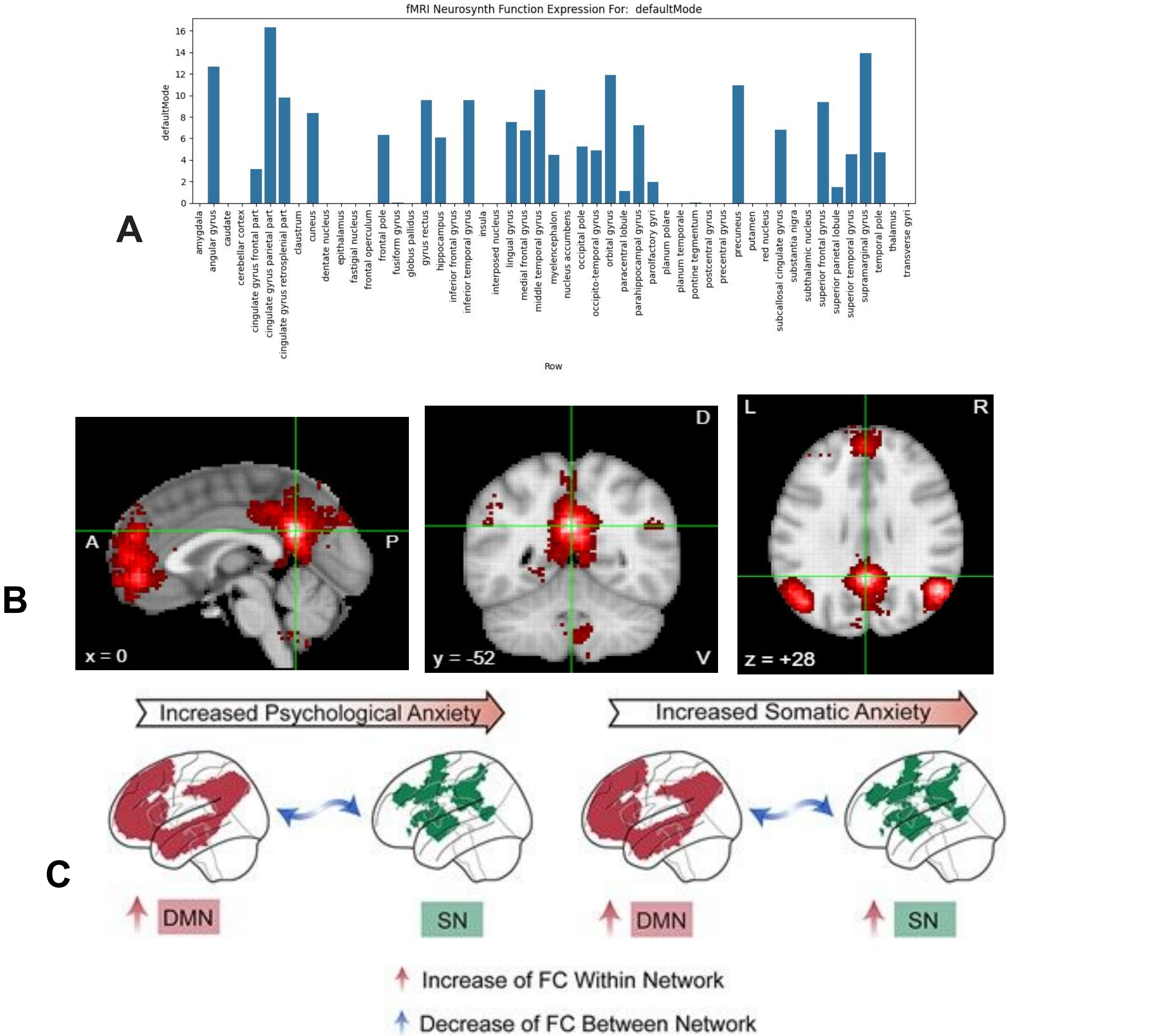


Figure 2. The default mode network (DMN) mapped and its role in generalized anxiety disorder (GAD).

(A) The fMRI neurosynth expression bar graph for the default mode network (DMN) is shown. The regions with the highest expression are the parietal part of the cingulate gyrus, the supramarginal gyrus, angular gyrus, orbital gyrus, and precuneus among others. **(B)** fMRI scans for the DMN neurosynth condition are shown. The areas shown in red are most active during rest/default. Notable areas that have high BOLD activity include the frontal cortex, cingulate cortex, angular gyrus, and precuneus. These areas work together to help coordinate many functions from mind-wandering to memory consolidation. **(C)** Using resting-state functional connectivity (FC) to investigate differences between GAD patients and healthy controls (HC), a study found significant differences in the DMN and its connectivity to the salience network (SN). GAD's psychological anxiety symptoms is associated with higher FC within the DMN, and the somatic anxiety dimension is associated with an increase of FC within both the DMN and SN. Additionally, both symptoms were associated with decreased FC between SN and DMN networks (Li et al., 2023). This data further supports how an overly connected DMN plays a role in excessive self reflection/rumination, and can be responsible for other psychological disorders as discussed in **Subsection 1 Figure 1**.

Subsection 3: Default Mode Network Receptor Analysis

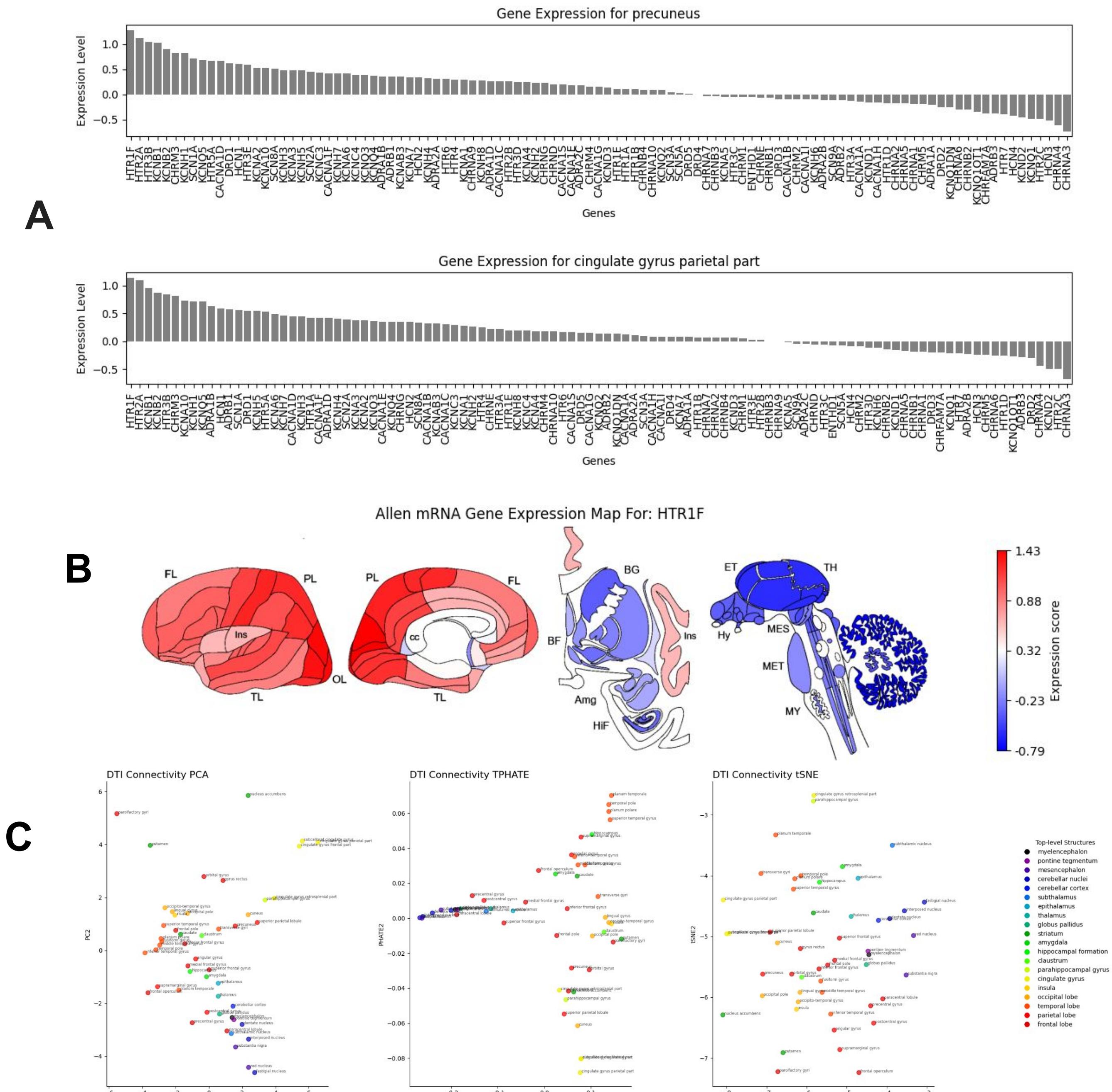


Figure 3. The 5-HT1F receptor is heavily expressed in brain regions that support the DMN.

(A) The gene expression for the precuneus and cingulate gyrus (parietal part) are shown due to their high involvement in the DMN as discussed in **Subsection 1 Figure 2**. HTR1F is the most expressed gene in both of these regions. The HTR1F gene encodes the 5-Hydroxytryptamine Receptor 1F which primarily binds to serotonin. **(B)** The expression map for HTR1F is shown. While this receptor is expressed in many regions of the cortex, it is also expressed in regions deeper within the brain that are known to be involved with the DMN, like the insula. Several receptor types play roles in modulating the DMN, including other serotonin receptor subtypes, dopamine D2 receptors (D2R), and nicotinic acetylcholine receptors (nAChR) (Froudist-Walsh et al., 2023) (Hahn et al., 2021). While the exact receptor to function connection is not completely understood, some experiments have found that certain receptor inhibition influences parts of the DMN. Inhibition of the 5-HT1A receptor influences memory-based future envisioning, and self-referential judgements. Both of which are known functions of the DMN (Hahn et al., 2012). **(C)** DTI connectivity is shown. Areas involved in the DMN and similar in gene expression, like the precuneus and cingulate gyrus are not highly correlated on the DTI connectivity chart. This is plausible, as these regions have differing functions outside of the DMN, which already has a wide range of functions.

Section 3: Receptor Report

Subsection 0: In Section 3, I will investigate the serotonin 5-HT2A receptor, detailing its molecular structure, widespread expression across the brain, and its significant role in modulating mood and cognitive functions. By analyzing gene expression data and functional studies, I aim to illustrate how the 5-HT2A receptor contributes to emotional regulation and its implications for psychiatric disorders such as depression and anxiety.

Subsection 1: The 5-HT2A Receptor

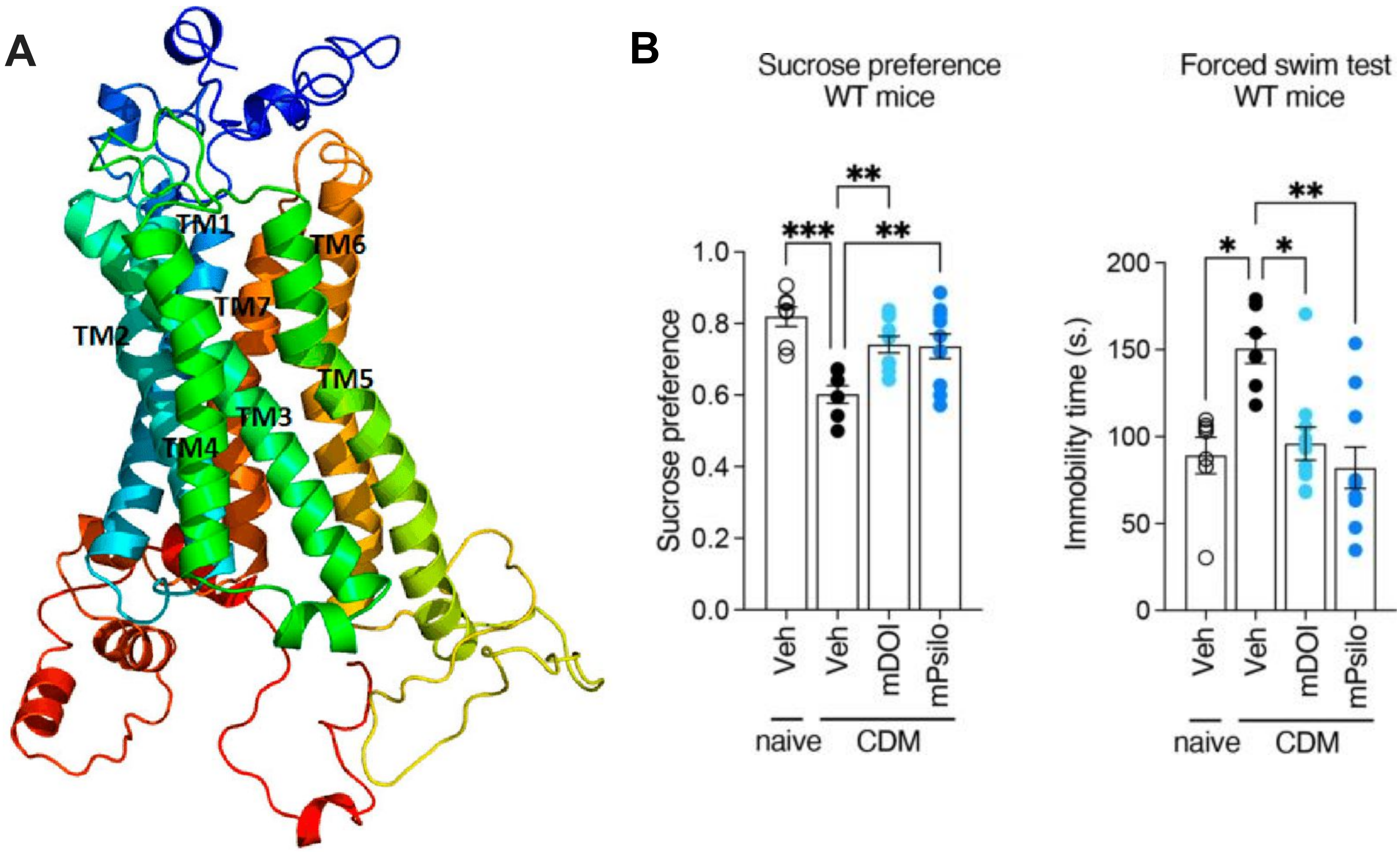
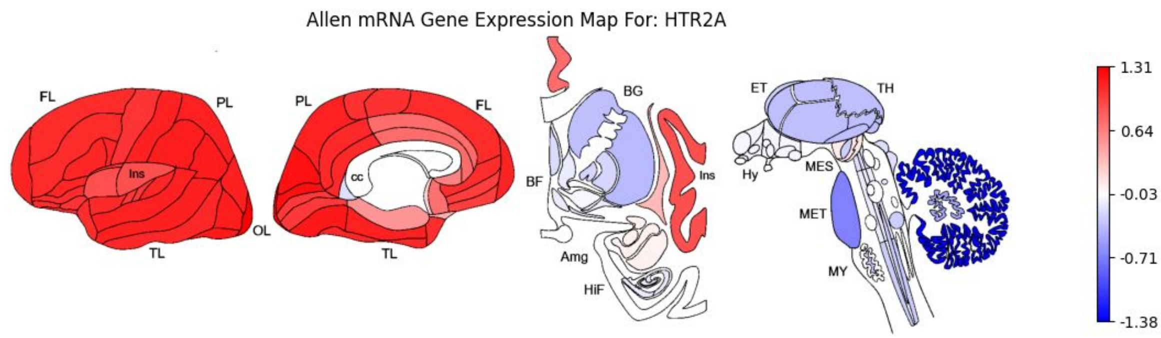


Figure 1. The 5-HT2A receptor is a subtype of serotonergic receptors with many functions including memory, cognition, and emotional processing. It plays a role in many psychological disorders.

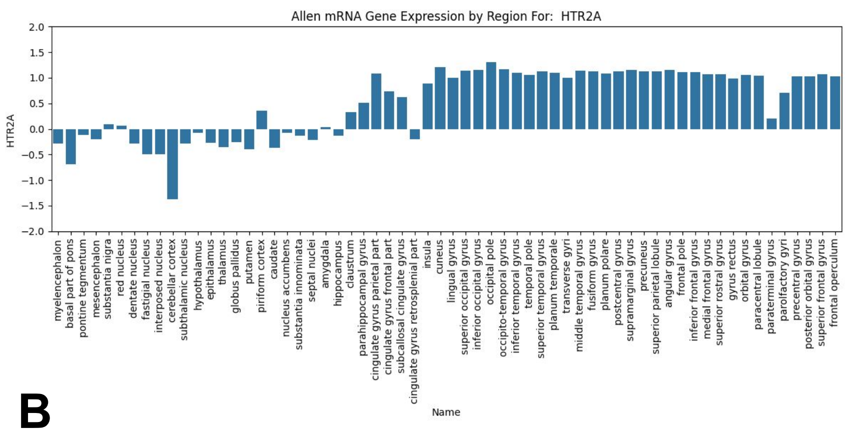
(A) Shown is the structure of the 5-HT2A receptor. It contains 7 transmembrane domains and 471 amino acids in humans. The domains are labeled in the picture TM1-TM7 (Sowdhamini et al., 2015). These receptors are widespread across the brain, and are involved in a variety of functions from mood to memory. The 2A variant is thought to play a role in mood and working memory, and many antidepressants and psychedelics act on the 5-HT2A receptor (Vargas et al., 2023). By understanding the function of this receptor, we can make better working and more targeted depression medications.

(B) Using a Sucrose Preference (SPT) and Forced Swimming test (FST) a study found that the microdosing of hallucinogenic 5-HT2A receptor agonists induces antidepressant-like effects in mice. The researchers investigated psilocybin and 2,5-Dimethoxy-4-iodoamphetamine (DOI) compared to vehicles/controls (Veh). This experiment supports the idea that specifically the 5-HT2A receptor plays a large role in mood, and explains why antidepressant medications target this receptor (Sekssaoui et al., 2024).

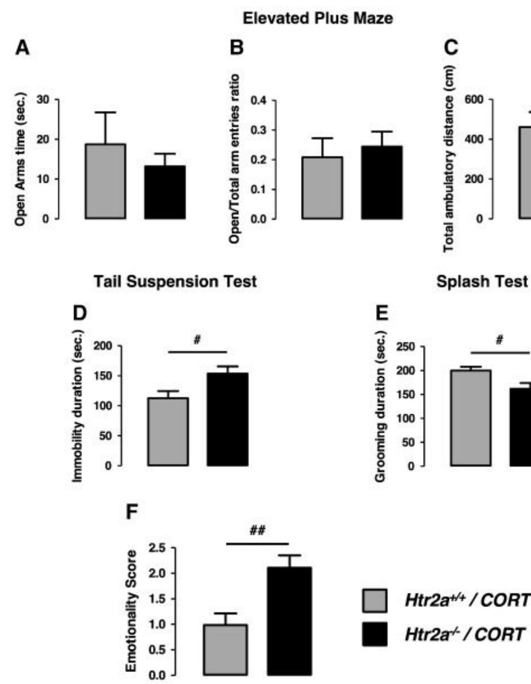
Subsection 2: 5-HT2A Receptor Expression in the Brain



A



B



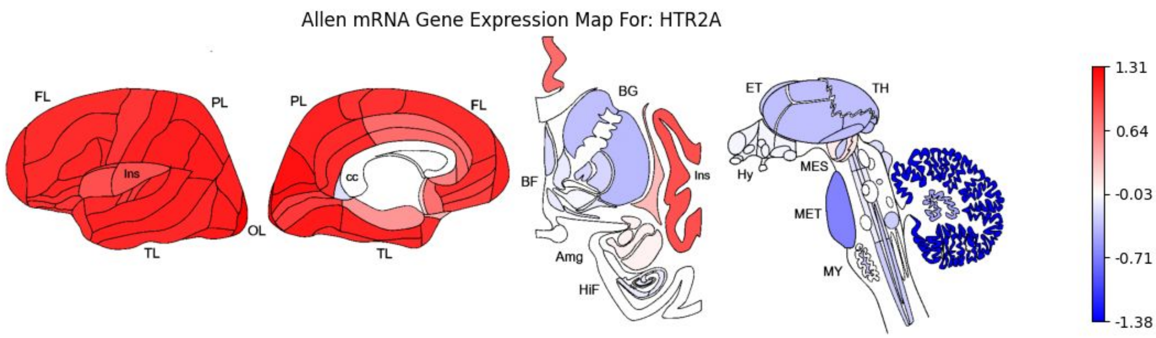
C

Figure 2. The 5-HT2A receptor is heavily expressed in most regions of the cortex, and is closely linked to anxiety and depressive disorders.

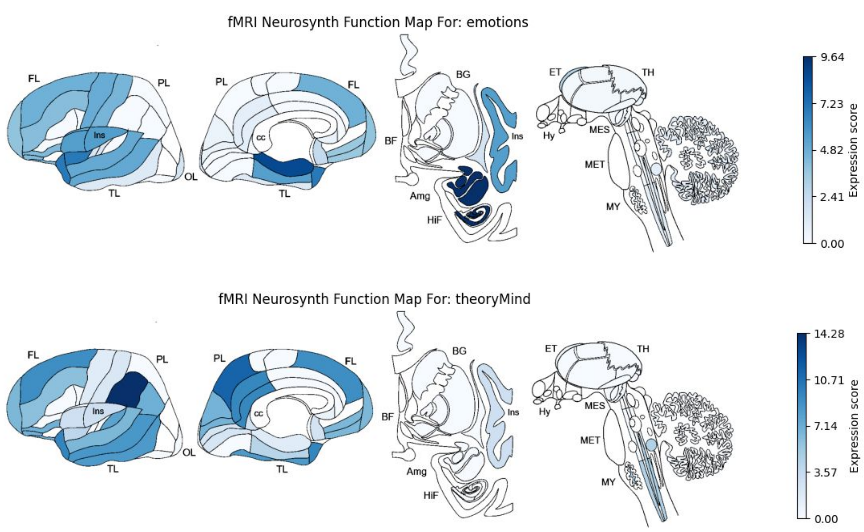
(A) The mRNA gene expression map for the 5-HT2A receptor is shown. It is expressed at high levels at nearly all regions of the cortex, and is suppressed the most in the cerebellar cortex. **(B)** Shown is the accompanying bar chart for the HTR2A gene. Here it is clear to see that HTR2A is expressed in many different regions, and is suppressed mainly by midbrain regions. **(C)** A study investigated behavioral differences after a corticosterone injection in HTR2A knockout mice using the Elevated Plus Maze (EPM), Tail Suspension (TST), and Splash test (ST). Anxiety is measured in the EPM, despair is measured in the TST, and self-care is measured in the ST. Overall, the HTR2A negative group was more prone to develop anxio-depressive-like responses in the different tests compared to controls (Petit et al., 2014). This experiment further demonstrates HTR2A's key role in mood and depressive symptoms.

Subsection 2: 5-HT2A Receptor Expression Compared to Neurosynth Functions

A



B



C

Type of Stress	Outcome	Reference
Studies in support of the hypothesis that stress increases 5-HT$_{2A}$ receptor expression or function		
Physical Stress		
Electroconvulsive shock	Increase in 5-HT$_2$ receptor binding sites and 5-HTP HTR	(Goodwin et al. 1984)
Electroconvulsive shock	Increase in 5-MeO-DMT HTR	(Metz and Heal 1986)
Acute or repeated immobilization	Increase in 5-HT$_2$ receptor binding sites	(Torda et al. 1988)
Chronic forced swimming	Increased DOI HTR and 5-HT$_2$ availability	(Takao et al. 1995)
Toe pinch stress	Alleviation of symptoms by DOI	(Hawkins et al. 2002)
Chronic unpredictable stress	Increased 5-HT$_{2A}$ expression	(McKittrick et al. 1995; Ossowska et al. 2001)
Social Stress		
Social isolation	Increased DOI HTR and 5-HT$_{2A}$ expression	(Takao et al. 1995; Ueki et al. 2015)
Social isolation	Increased DOI HTR	(Brotto et al. 1998)
Mixed sex social housing	Increase in 5-HT$_2$ receptor binding sites	(McKittrick et al. 1995)
Social defeat	Increased 5-HT$_2$ availability	(Berton et al. 1998)
Maternal Care Stress		
Maternal separation	Increased DOI HTR and altered c-Fos expression	(Sood et al. 2018)
Prenatal exposure to LPS	Increased DOI HTR and 5-HT$_{2A}$ expression	(Holloway et al. 2013; Wischhof et al. 2015)
Maternal variable stress	Increased DOI HTR and 5-HT$_{2A}$ expression	(Holloway et al. 2013; Ueki et al. 2015)
Studies not in support of the hypothesis that stress increases 5-HT$_{2A}$ receptor expression or function		
Adult exposure to LPS	Decreased HTR	(Kouhata et al. 2001)
Single footshock	Decreased HTR but unaltered 5-HT$_{2A}$ expression	(Izumi et al. 2002)
Acute forced swimming	Decreased DOI or 5-HTP HTR	(Pericic 2003)
Chronic unpredictable stress	5-HT$_{2A}$ knock mice show unaltered depression-like states	(Jaggar et al. 2017)

Figure 3. The 5-HT2A receptor is expressed in the regions that are responsible for mood and emotions, and is upregulated during periods of stress.

(A) The mRNA gene expression map for the 5-HT2A receptor is shown for comparison to the neurosynth data in (B). **(B)** Shown are fMRI neurosynth function maps for emotions and theoryMind. These regions were chosen due to their connection to mood and depressive disorders. There exists a fairly clear overlap between the HTR2A expression and these function maps. The function map for emotions matches particularly well due to its high insula activity. Overall, while there are differences, the expression map matches the function map well. This is to be expected, and is not surprising due to what is known about the function of the 5-HT2A receptor. **(C)** Shown is a table of many studies that have investigated how different types of stress impact 5-HT2A receptor expression. There is a clear consensus that different types of stress typically lead to increased 5-HT2A expression, even though a few studies do not support this hypothesis (Murnane, 2020).

References

Ballesteros, J. J., Briscoe, S. D., & Ishizawa, Y. (2020). Neural signatures of α2-adrenergic agonist-induced unconsciousness and awakening by antagonist. eLife, 9, e57685. https://doi.org/10.7554/eLife.57685

Bjerke, I. E., Yates, S. C., Laja, A., Witter, M. P., Puchades, M. A., Bjaalie, J. G., & Leergaard, T. B. (2021). Densities and numbers of calbindin and parvalbumin positive neurons across the rat and mouse brain. iScience, 24(1), 101906. https://doi.org/10.1016/j.isci.2020.101906

Crick, F., & Koch, C. (2003). A framework for consciousness. Nature Neuroscience, 6(2), 119–126. https://doi.org/10.1038/nn0203-119

Duffy, K. A., Rosch, K. S., Nebel, M. B., Seymour, K. E., Lindquist, M. A., Pekar, J. J., Mostofsky, S. H., & Cohen, J. R. (2021). Increased integration between default mode and task-relevant networks in children with ADHD is associated with impaired response control. Developmental Cognitive Neuroscience, 50, 100980. https://doi.org/10.1016/j.dcn.2021.100980

Froudist-Walsh, S., Browning, P. G., Young, J. J., Murphy, K. L., Mars, R. B., Fuzat, E. C., & Croxson, P. L. (2023). Chemogenetic inhibition of the primate cortico-claustral circuit alters functional connectivity. Nature Neuroscience, 26(8), 1394–1402. https://doi.org/10.1038/s41593-023-01351-2

GeneCards. (n.d.). ADRA2C Gene - GeneCards | ADA2C Protein | ADA2C Antibody. Retrieved April 20, 2025, from https://www.genecards.org/cgi-bin/carddisp.pl?gene=ADRA2C

Hahn, B., Harvey, A. N., Concheiro-Guisan, M., Huestis, M. A., Ross, T. J., & Stein, E. A. (2021). Nicotinic receptor modulation of the default mode network. Psychopharmacology, 238(2), 589–597. https://doi.org/10.1007/s00213-020-05711-9

Hahn, A., Stein, P., Windischberger, C., Weissenbacher, A., Spindelegger, C., Moser, E., Kasper, S., & Lanzenberger, R. (2012). Reduced resting-state functional connectivity between amygdala and orbitofrontal cortex in social anxiety disorder. NeuroImage, 56(3), 881–889. https://doi.org/10.1016/j.neuroimage.2011.02.064

Hein, L., Altman, J. D., & Kobilka, B. K. (1999). Two functionally distinct α2-adrenergic receptors regulate sympathetic neurotransmission. Nature, 402(6758), 181–184. https://doi.org/10.1038/46040

International Union of Basic and Clinical Pharmacology. (n.d.). α2C-adrenoceptor. Retrieved April 19, 2025, from https://www.guidetopharmacology.org/GRAC/ObjectDisplayForward?objectId=27

Li, W., Qin, W., Liu, H., Fan, L., Wang, J., Jiang, T., & Yu, C. (2013). Subregions of the human superior frontal gyrus and their connections. NeuroImage, 78, 46–58. https://doi.org/10.1016/j.neuroimage.2013.04.011

Madden, M. B., Stewart, B. W., White, M. G., Krimmel, S. R., Wang, F., Mathur, B. N., & McDannald, M. A. (2022). Claustrum parvalbumin interneurons mediate feedforward inhibition of the anterior cingulate cortex. Cerebral Cortex, 33(10), 6354–6367. https://doi.org/10.1093/cercor/bhac508

Online Mendelian Inheritance in Man. (n.d.). #104250 - ALPHA-2C-ADRENERGIC RECEPTOR; ADRA2C. Retrieved April 19, 2025, from https://omim.org/entry/104250

Petit, A.-C., Quesseveur, G., Gressier, F., Colle, R., David, D. J., Gardier, A. M., Ferreri, F., Lepine, J.-P., Falissard, B., Verstuyft, C., Beaupré, F., & Corruble, E. (2014). Converging translational evidence for the involvement of the serotonin 2A receptor gene in major depressive disorder. Progress in Neuro-Psychopharmacology and Biological Psychiatry, 54, 76–82. https://doi.org/10.1016/j.pnpbp.2014.04.013

Torgerson, C. M., & Van Horn, J. D. (2014). A case study in connectomics: The history, mapping, and connectivity of the claustrum. Frontiers in Neuroinformatics, 8, 83. https://doi.org/10.3389/fninf.2014.00083

Torgerson, C. M., Irimia, A., Goh, S. Y. M., & Van Horn, J. D. (2015). The DTI connectivity of the human claustrum. Human Brain Mapping, 36(3), 827–838. https://doi.org/10.1002/hbm.22667

Vargas, M. V., Dunlap, L. E., Dong, C., Carter, S. J., Tombari, R. J., Jami, S. A., Cameron, L. P., Patel, S. D., Hennessey, J. J., Safron, A., Allen, J. R., Bathini, S., Borden, P., Bawmann, V., Bertozzi, C. R., Carhart-Harris, R., Carter, O. L., Dunlap, L. E., & Olson, D. E. (2023). Psychedelics promote neuroplasticity through the activation of intracellular 5-HT2A receptors. Science, 379(6633), 700–706. https://doi.org/10.1126/science.adf0435

Gandhimathi, A., & Sowdhamini, R. (2016). Molecular modelling of human 5-hydroxytryptamine receptor (5-HT2A) and virtual screening studies towards the identification of agonist and antagonist molecules. Journal of Biomolecular Structure and Dynamics, 34(5), 952–970. https://doi.org/10.1080/07391102.2015.1062802

Sekssaoui, M., Bockaert, J., Marin, P., & Bécamel, C. (2024). Antidepressant-like effects of psychedelics in a chronic despair mouse model: Is the 5-HT2A receptor the unique player? Neuropsychopharmacology, 49(4), 747–756. https://doi.org/10.1038/s41386-024-01794-6

Rodríguez-Vidal, L., Alcauter, S., & Barrios, F. A. (2024). The functional connectivity of the human claustrum, according to the Human Connectome Project database. PLOS ONE, 19(4), Article e0298349. https://doi.org/10.1371/journal.pone.0298349

Murnane, K. S. (2019). Serotonin 2A receptors are a stress response system: Implications for post-traumatic stress disorder. Behavioural Pharmacology, 30(1), 1–13. https://doi.org/10.1097/FBP.0000000000000459

Li, R., Shen, F., Sun, X., Zou, T., Li, L., Wang, X., Deng, C., Duan, X., He, Z., Yang, M., Li, Z., & Chen, H. (2023). Dissociable salience and default mode network modulation in generalized anxiety disorder: A connectome-wide association study. Cerebral Cortex, 33(10), 6354–6365. https://doi.org/10.1093/cercor/bhac509